

REPOMIND

AI Software Reliability System

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Ask Before You Break

IBM Bob 2.0 Hackathon

TEAM



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AI/ML Engineer & Full-Stack Developer



PROBLEM



- AI-generated bug findings are hypotheses
- Static reasoning isn't runtime evidence
- Generated tests can themselves be wrong

RESEARCH QUESTION

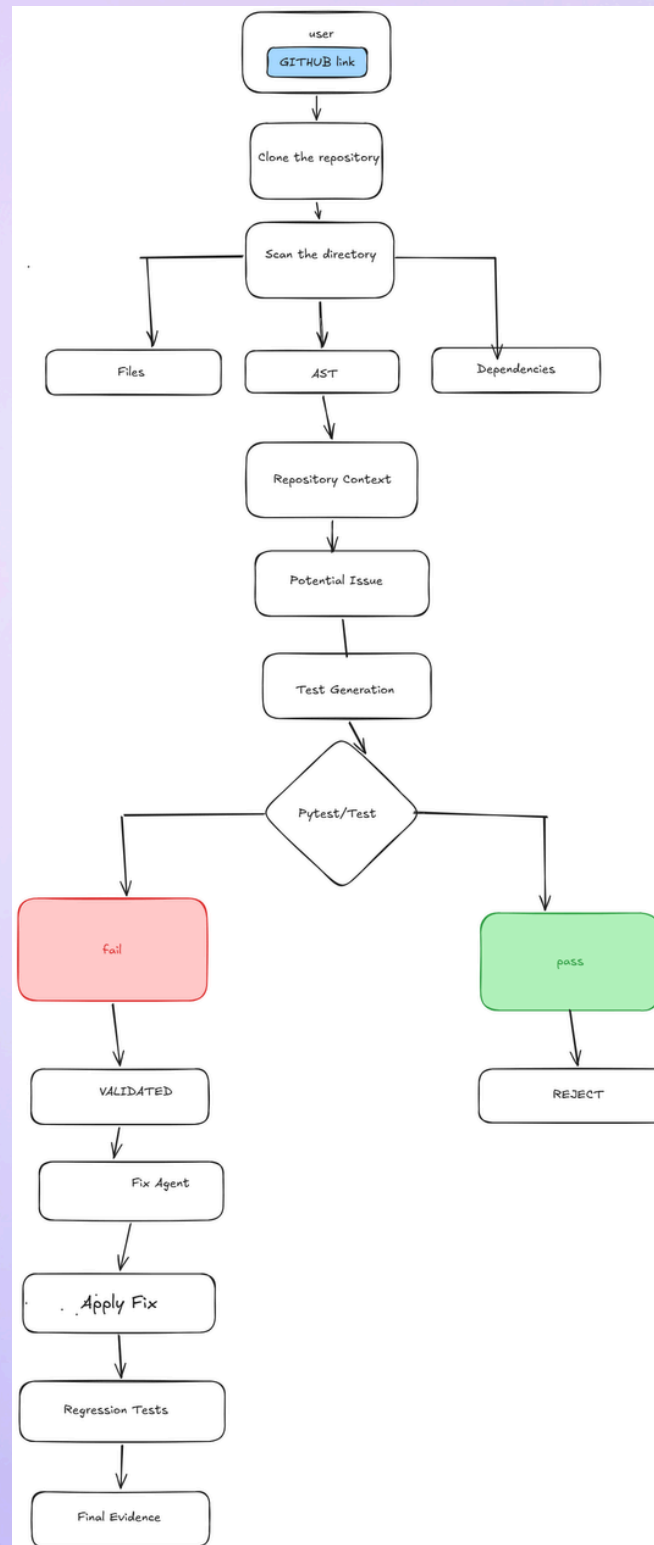
Can an AI agent autonomously discover and validate software reliability issues in an unfamiliar codebase using repository-level context and iterative experimentation?

Solution

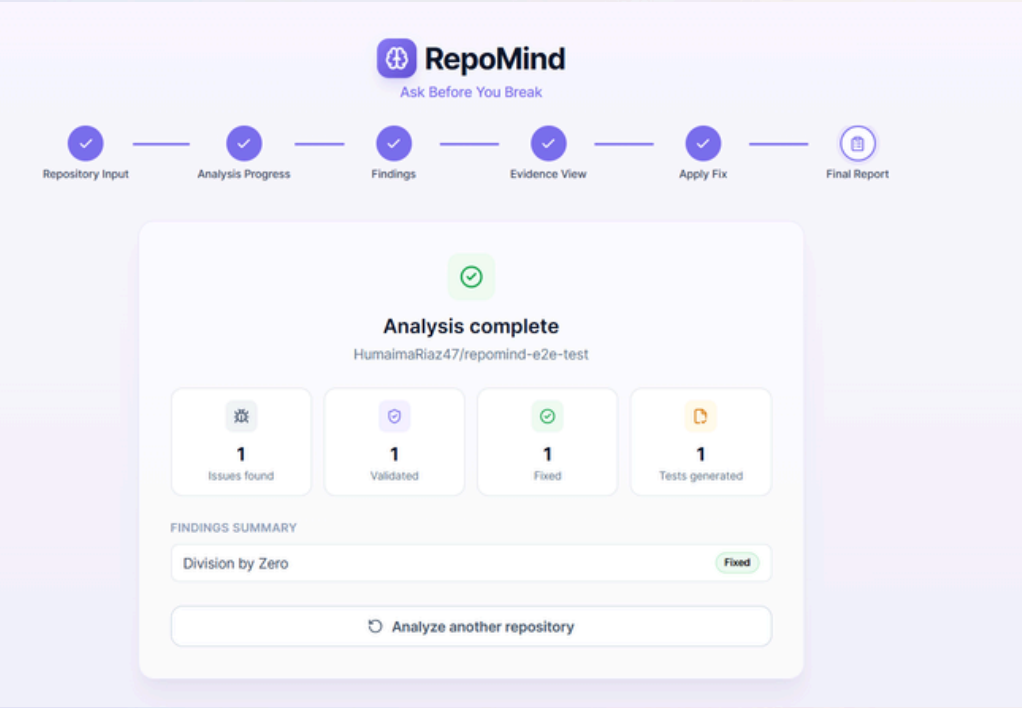
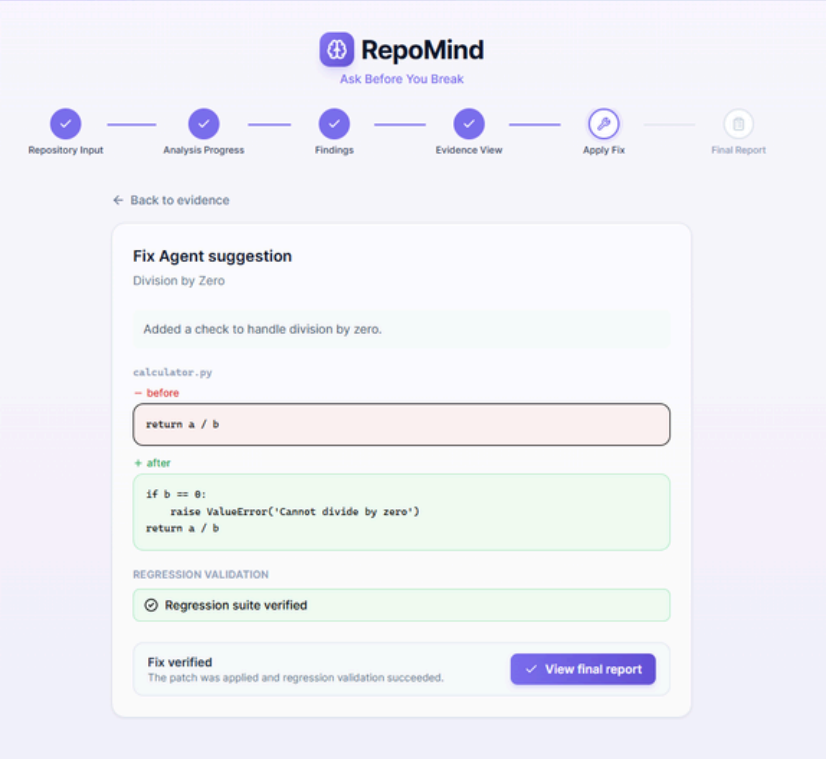
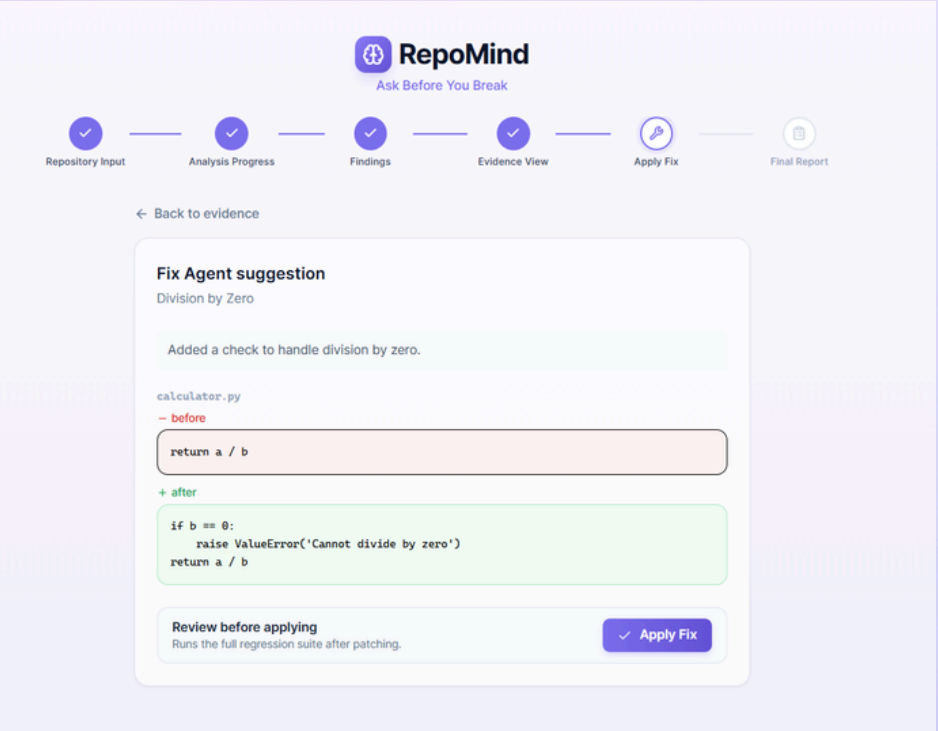
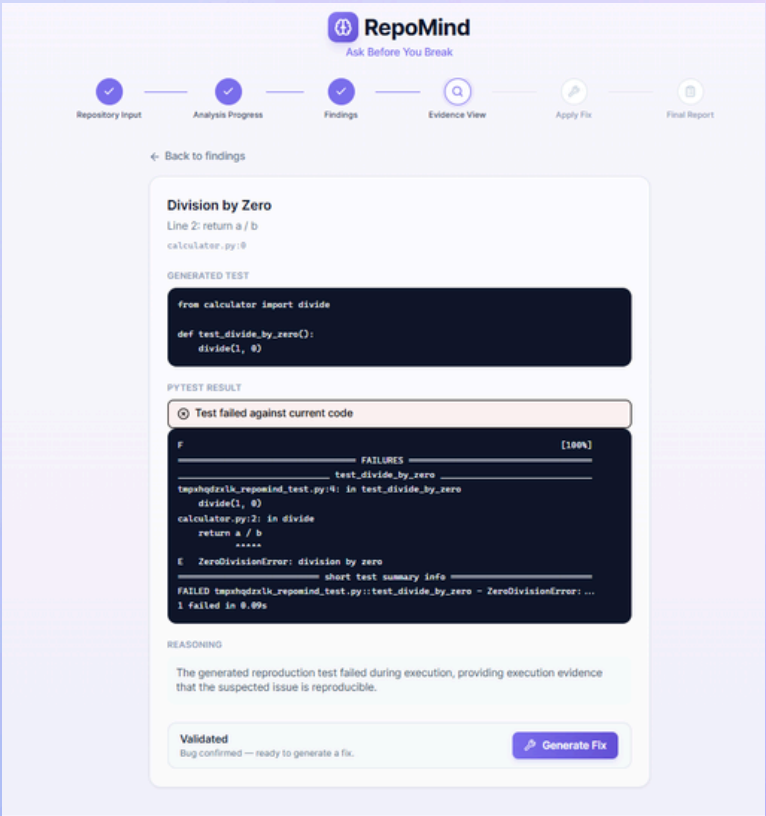
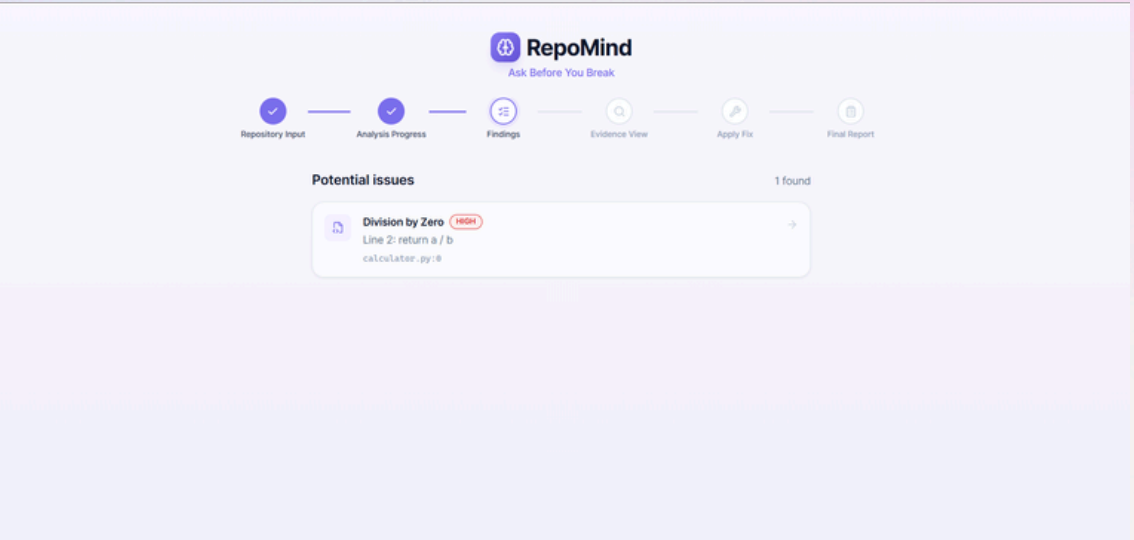
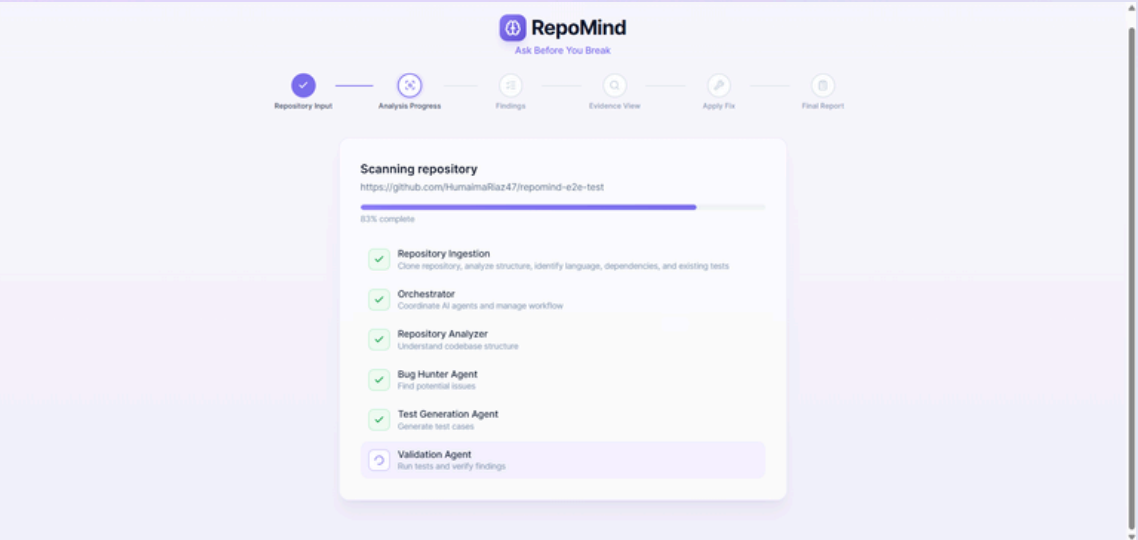
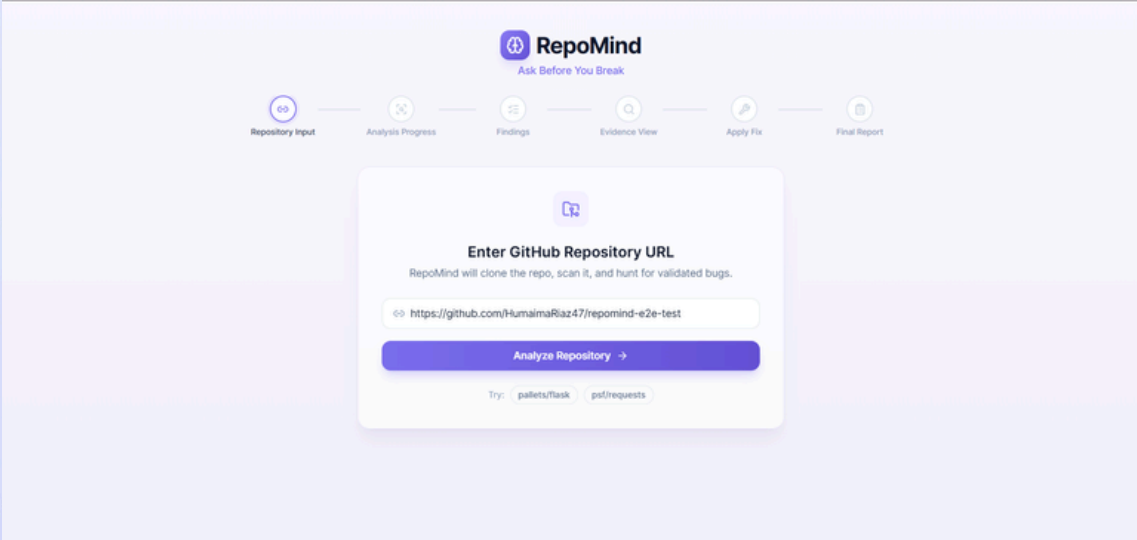
Analyze → Detect → Generate Test → Execute → Validate → Fix → Regression



ARCHITECTURE

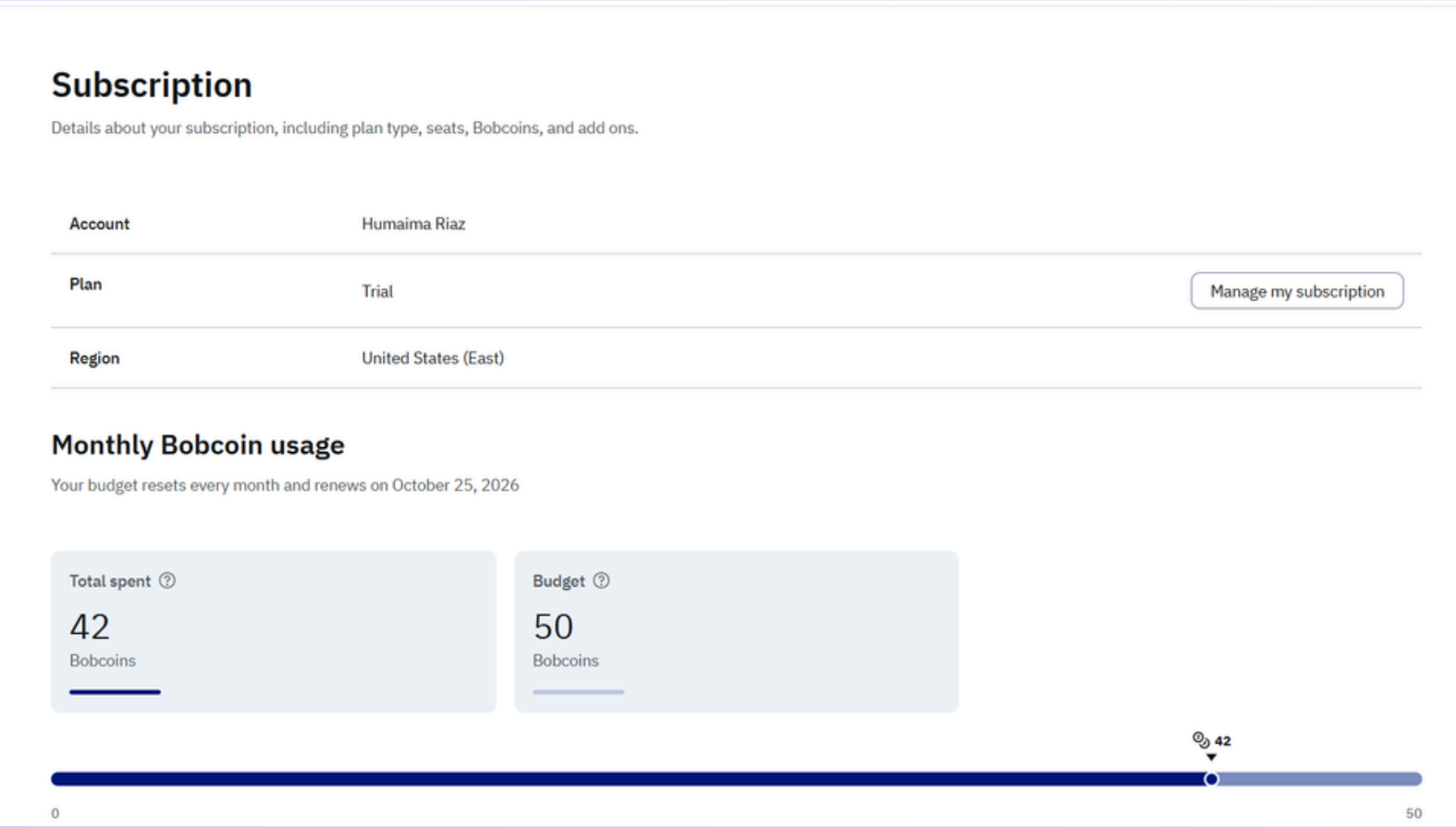
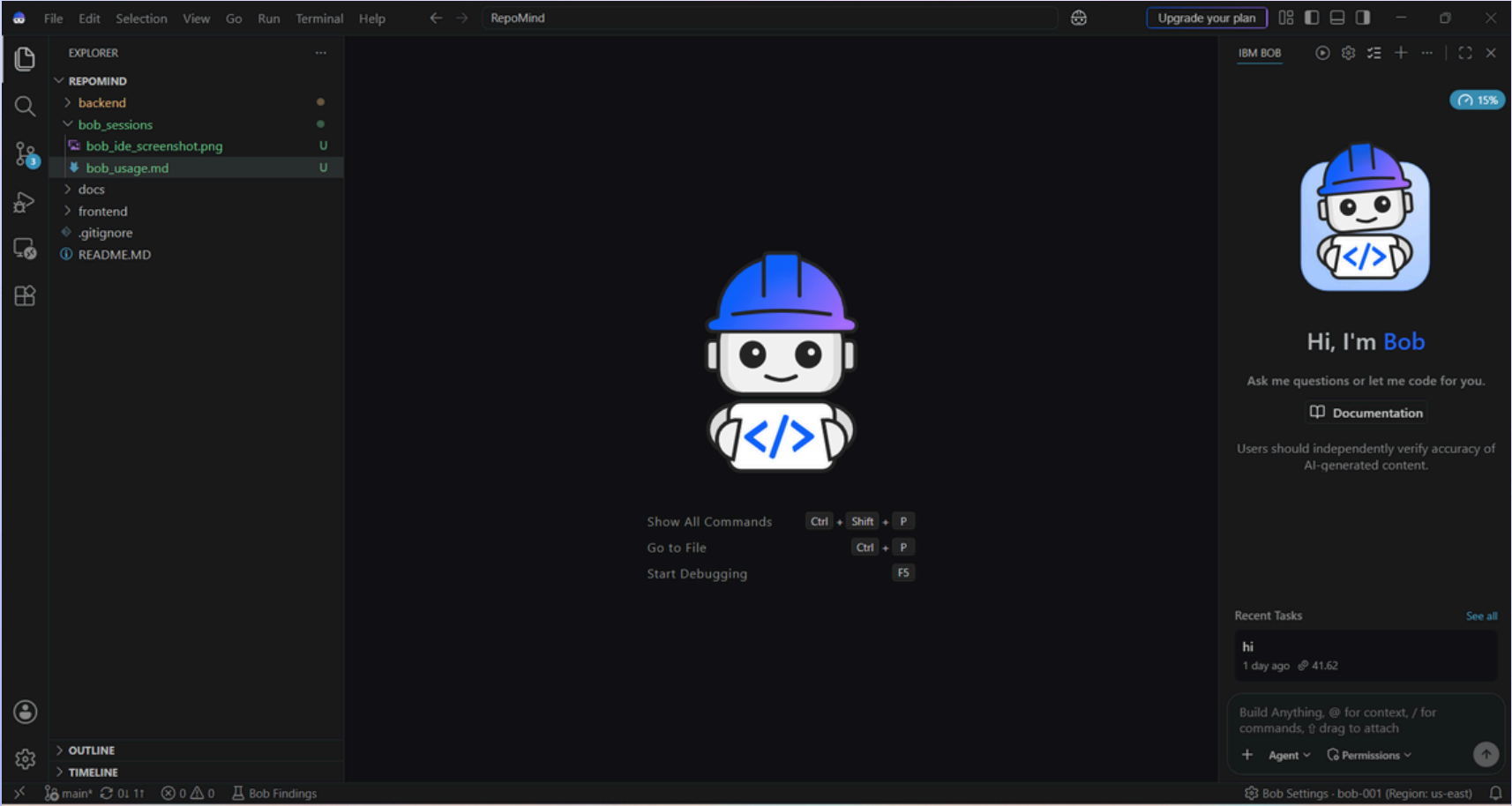


LIVE WORKFLOW



IBM BOB USAGE

Bob used throughout development - 42 / 50 Bobcoins used - Major Bob-assisted development tasks



IMPACT

RepoMind shifts AI-assisted software engineering from “AI says it is a bug” to “AI found hypothesis, and the code proved it.”

- Evidence-based bug validation
- Faster debugging and testing
- Fewer false AI conclusions
- Safer AI-generated fixes

FUTURE

An autonomous software reliability layer that continuously tests AI-generated hypotheses against real software behavior.

- CI/CD & GitHub PR integration
- Multi-language support
- Security vulnerability detection
- Continuous repository monitoring

× **THANK YOU** ×

RepoMind — AI Software Reliability & Validation System

“Ask Before You Break.”

Analyze → Test → Execute → Validate → Fix → Verify